

Curriculum for B.Sc. in Computer Science and Engineering

Version: Summer-2018

Curriculum Name: B.Sc. in Computer Science and Engineering

Minimum Requirement 140 Credits

Credits Requirement for the degree B.Sc. in Computer Science and Engineering

SL	Name	Credits
1	COMPULSORY LANGUAGE AND GENERAL EDUCATION COURSES	9
2	ELECTIVE GENERAL EDUCATION COURSES	9
3	BASIC SCIENCE COURSES	11
4	MATHEMATICS AND STATISTICS COURSES	15
5	CORE COMPUTER SCIENCE AND ENGINEERING COURSES	62
6	CAPSTONE PROJECT	6
7	NON-MAJOR ELECTIVE COMPUTER SCIENCE AND ENGINEERING COURSES	8
8	MAJOR	20
Total Credits		140

Compulsory Language and General Education Courses **(9 credits)**

SL	Course Code	Course Name	Credits
1	ENG101 / ENG7101	Basic English / Basic English	3
2	ENG102 / ENG7102	Composition and Communication Skills / Composition and Communication Skills	3
3	GEN226 / GEN7226	Emergence of Bangladesh / Emergence of Bangladesh	3

Elective General Education Courses **(9 credits)**

SL	Course Code	Course Name	Credits
1	ACT101	Financial Accounting	3
2	BUS231	Business Communication	3
3	BUS321	Business for Engineering and Technology	3

SL	Course Code	Course Name	Credits
4	ECO101 / ECO7101	Principles of Microeconomics / Principles of Microeconomics	3
5	FIN101	Principles of Finance	3
6	GEN201	Bangladesh Studies	3
7	GEN202	Eastern Culture and Heritage	3
8	GEN203	Ecological System and Environment	3
9	GEN204	Western Thought	3
10	GEN205	Introduction to Psychology	3
11	GEN207	Industrial Psychology	3
12	GEN208	Introduction to Philosophy	3
13	GEN209	Social Psychology	3
14	GEN210	International Relations	3
15	GEN211	Concepts of Journalism & Media Studies	3
16	GEN214	Introduction to Development Studies	3
17	GEN224	Bengali Language	3
18	GEN239	Professional Ethics	3
19	GEN7201	Bangladesh Studies	3
20	GEN7202	Eastern Culture and Heritage	3
21	GEN7203 / GEN7226	Ecological System and Environment / Emergence of Bangladesh	3
22	GEN7204	Western Thought	3
23	GEN7205	Introduction to Psychology	3
24	GEN7207	Industrial Psychology	3
25	GEN7208	Introduction to Philosophy	3
26	GEN7209	Introduction to Social Psychology	3
27	GEN7210	International Relation	3
28	GEN7211	Concepts of Journalism & Media Studies	3
29	GEN7213 / GEN213	Introduction to German Language / Introduction to German Language	3
30	GEN7214	Introduction to Development Studies	3
31	GEN7215 / GEN215	Introduction to French Language / Introduction to French Language	3

SL	Course Code	Course Name	Credits
32	GEN7216 / GEN216	Introduction to Spanish Language / Introduction to Spanish Language	3
33	GEN7217 / GEN217	Introduction to Chinese Language / Introduction to Chinese Language	3
34	GEN7218 / GEN218	Introduction to Arabic Language / Introduction to Arabic Language	3
35	GEN7220 / GEN220	Principles of Public Relations / Principles of Public Relations	3
36	GEN7221 / GEN221	Globalization, Development and Change / Globalization, Development and Change	3
37	GEN7223 / GEN223	Contemporary Security Studies in Asia-Pacific / Contemporary Security Studies in Asia-Pacific	3
38	GEN7224	Bangla Language	3
39	GEN7239	Professional Ethics	3
40	MGT321	Industrial Management	3
41	MGT337	Production Operations Management	3
42	MKT101	Principles of Marketing	3
43	POP201	Health Challenges of Adolescents and Youths	3
44	POP202	Introduction to Public Health	3
45	POP203	Introduction to Population Studies	3
46	POP7201	Health Challenges of Adolescents and Youth	3
47	POP7202	Introduction to Public Health	3
48	POP7203	Introduction to Population Studies	3
49	SOC211	Eastern Culture and Heritage	3
50	SOC217	Religion, Ethnicity, Culture and Development in South Asia	3
51	SOC317	Sociology of Science and Technology	3
52	SOC7217	Religion, Ethnicity, Culture and Development in South Asia	3

Basic Science Courses

(11 credits)

SL	Course Code	Course Name	Credits
1	CHE109	Engineering Chemistry-I	4
2	PHY109	Engineering Physics-I	4

SL	Course Code	Course Name	Credits
3	PHY209	Engineering Physics-II	3

Mathematics and Statistics Courses**(15 credits)**

SL	Course Code	Course Name	Credits
1	MAT101	Differential & Integral Calculus	3
2	MAT102	Differential Equations & Special Functions	3
3	MAT104	Co-ordinate Geometry & Vector Analysis	3
4	MAT205	Linear Algebra & Complex Variables	3
5	STA102	Statistics and Probability	3

Core Computer Science and Engineering Courses**(62 credits)**

SL	Course Code	Course Name	Credits
1	CSE103	Structured Programming	4.5
2	CSE106	Discrete Mathematics	3
3	CSE110	Object Oriented Programming	4.5
4	CSE200	Computer-Aided Engineering Drawing	1
5	CSE207	Data Structures	4
6	CSE209	Electrical Circuits	4
7	CSE246	Algorithms	4.5
8	CSE251	Electronic Circuits	4
9	CSE302	Database Systems	4.5
10	CSE325	Operating Systems	4
11	CSE345	Digital Logic Design	4
12	CSE347	Information System Analysis and Design	4
13	CSE360	Computer Architecture	3
14	CSE405	Computer Networks	4
15	CSE407	Green Computing	3
16	CSE487	Computer and Cyber Security	3
17	CSE495	IT Project Management and Entrepreneurship	3

Capstone Project**(6 credits)**

Completed at least 105 credit hours

SL	Course Code	Course Name	Credits
1	CSE400A	Capstone Project (Part 1 of 3)	1
2	CSE400B	Capstone Project (Part 2 of 3)	2
3	CSE400C	Capstone Project (Part 3 of 3)	3

Non-major Elective Computer Science and Engineering Courses (8 credits)

Minimum 8 Credits (two to three courses depending on credits of the courses) from one or more major/non-major areas other than selected major area

SL	Course Code	Course Name	Credits
1	CSE225	Numerical Methods	4
2	CSE303	Statistics for Data Science	4
3	CSE313	Theory of Computations	3
4	CSE350	Data Communications	4
5	CSE355	Digital System Design	4
6	CSE366	Artificial Intelligence	4
7	CSE406	Internet of Things	4
8	CSE412	Software Engineering	4
9	CSE420	Computer Graphics	4
10	CSE422	Simulation and Modeling	4
11	CSE423	Software Architecture	4
12	CSE428	Human Computer Interactions	4
13	CSE430	Software Testing and Quality Assurance	4
14	CSE432	Digital Signal Processing	4
15	CSE438	Digital Image Processing	4
16	CSE442	Microprocessors and Microcontrollers	4
17	CSE445	Computer Vision	4
18	CSE446	ASIC Design Using FPGA	4
19	CSE452	Distributed Systems and Algorithms	4
20	CSE453	Wireless Network	4
21	CSE457	Cellular Networks	4
22	CSE460	Cryptography	3

SL	Course Code	Course Name	Credits
23	CSE464	Advanced Database System	4
24	CSE471	Compiler Design	4
25	CSE472	Advanced Network Services and Management	4
26	CSE473	Network Security and Systems	4
27	CSE474	Pattern Recognition	4
28	CSE475	Machine Learning	4
29	CSE477	Data Mining	4
30	CSE479	Web Programming	4
31	CSE481	Nature-Inspired Computing	4
32	CSE483	Graph Theory	3
33	CSE484	Computational Geometry	3
34	CSE486	Bioinformatics Algorithms	4
35	CSE488	Big Data Analytics	4
36	CSE489	Mobile Programming	4
37	CSE491	VLSI Design	4
38	CSE492	Robotics	4
39	CSE494	Embedded Systems	4

Major Area Courses

Communications and Networking

(20 credits)

SL	Course Code	Course Name	Credits
1	CSE350	Data Communications	4
2	CSE432	Digital Signal Processing	4
3	CSE452	Distributed Systems and Algorithms	4
4	CSE453	Wireless Network	4
5	CSE457	Cellular Networks	4
6	CSE489	Mobile Programming	4

Software Engineering

(20 credits)

SL	Course Code	Course Name	Credits
1	CSE412	Software Engineering	4
2	CSE422	Simulation and Modeling	4
3	CSE423	Software Architecture	4
4	CSE428	Human Computer Interactions	4
5	CSE430	Software Testing and Quality Assurance	4
6	CSE452	Distributed Systems and Algorithms	4
7	CSE464	Advanced Database System	4
8	CSE479	Web Programming	4
9	CSE489	Mobile Programming	4

Hardware Engineering

(20 credits)

SL	Course Code	Course Name	Credits
1	CSE355	Digital System Design	4
2	CSE406	Internet of Things	4
3	CSE442	Microprocessors and Microcontrollers	4
4	CSE491	VLSI Design	4
5	CSE492	Robotics	4
6	CSE494	Embedded Systems	4

Intelligence Systems and Data Science

(20 credits)

SL	Course Code	Course Name	Credits
1	CSE303	Statistics for Data Science	4
2	CSE366	Artificial Intelligence	4
3	CSE420	Computer Graphics	4
4	CSE438	Digital Image Processing	4
5	CSE452	Distributed Systems and Algorithms	4
6	CSE474	Pattern Recognition	4
7	CSE477	Data Mining	4
8	CSE486	Bioinformatics Algorithms	4

Equivalent Courses

SL	Course Name	Equi. Course Name	Credits
1	CSE209: Electrical Circuits	EEE101: Electrical Circuits I	4
2	ECO101: Principles of Microeconomics	ECO7101: Principles of Microeconomics	3
3	ECO102: Introduction to Macroeconomics	ECO7102: Introduction to Macroeconomics	3
4	ENG101: Basic English	ENG7101: Basic English	3
5	ENG102: Composition and Communication Skills	ENG7102: Composition and Communication Skills	3
6	GEN202: Eastern Culture and Heritage	GEN7202: Eastern Culture and Heritage	3
7	GEN226: Emergence of Bangladesh	GEN7226: Emergence of Bangladesh	3
Total Credits			22

Course Description

ACT101: Financial Accounting

Credits: 3

Course Description:

Prerequisites:

BUS231: Business Communication

Credits: 3

Course Description:

Prerequisites: ENG7101

BUS321: Business for Engineering and Technology

Credits: 3

Course Description:

Prerequisites: ENG7101

CHE109: Engineering Chemistry-I

Credits: 4

Course Description:

Prerequisites:

CSE103: Structured Programming

Credits: 4.5

Course Description:

Prerequisites:

CSE106: Discrete Mathematics

Credits: 3

Course Description:

Prerequisites: CSE103

CSE110: Object Oriented Programming

Credits: 4.5

Course Description:

Prerequisites: CSE103

CSE200: Computer-Aided Engineering Drawing

Credits: 1

Course Description:

Prerequisites:

CSE207: Data Structures

Credits: 4

Course Description:

Prerequisites: CSE110

CSE209: Electrical Circuits

Credits: 4

Course Description:

Prerequisites:

CSE225: Numerical Methods

Credits: 4

Course Description:

Prerequisites: CSE103

CSE246: Algorithms

Credits: 4.5

Course Description:

Prerequisites: CSE207

CSE251: Electronic Circuits

Credits: 4

Course Description:

Prerequisites: CSE209

CSE302: Database Systems

Credits: 4.5

Course Description:

Prerequisites: CSE106

CSE303: Statistics for Data Science

Credits: 4

Course Description:

Prerequisites: STA102

CSE313: Theory of Computations

Credits: 3

Course Description:

Prerequisites: CSE246

CSE325: Operating Systems

Credits: 4

Course Description:

Prerequisites: CSE207

CSE345: Digital Logic Design

Credits: 4

Course Description:

Prerequisites: CSE251

CSE347: Information System Analysis and Design

Credits: 4

Course Description:

Prerequisites: CSE302

CSE350: Data Communications

Credits: 4

Course Description:

Prerequisites: CSE251

CSE355: Digital System Design

Credits: 4

Course Description:

Prerequisites: CSE345

CSE360: Computer Architecture

Credits: 3

Course Description:

Prerequisites: CSE325

CSE366: Artificial Intelligence

Credits: 4

Course Description:

Prerequisites: CSE246

CSE400: Capstone Project

Credits: 6

Course Description:

Prerequisites:

CSE400A: Capstone Project (Part 1 of 3)

Credits: 1

Course Description:

Prerequisites:

CSE400B: Capstone Project (Part 2 of 3)

Credits: 2

Course Description:

Prerequisites: CSE400A

CSE400C: Capstone Project (Part 3 of 3)

Credits: 3

Course Description:

Prerequisites: CSE400B

CSE405: Computer Networks

Credits: 4

Course Description:

Prerequisites: CSE246

CSE406: Internet of Things

Credits: 4

Course Description:

Prerequisites: CSE405

CSE407: Green Computing

Credits: 3

Course Description:

Prerequisites: CSE405

CSE412: Software Engineering

Credits: 4

Course Description:

Prerequisites: CSE347

CSE420: Computer Graphics

Credits: 4

Course Description:

Prerequisites: CSE246

CSE422: Simulation and Modeling

Credits: 4

Course Description:

Prerequisites: CSE246

CSE423: Software Architecture

Credits: 4

Course Description:

Prerequisites: CSE412

CSE428: Human Computer Interactions

Credits: 4

Course Description:

Prerequisites: CSE412

CSE430: Software Testing and Quality Assurance

Credits: 4

Course Description:

Prerequisites: CSE412

CSE432: Digital Signal Processing

Credits: 4

Course Description:

Prerequisites: CSE246,CSE347

CSE438: Digital Image Processing

Credits: 4

Course Description:

Prerequisites: CSE246

CSE442: Microprocessors and Microcontrollers

Credits: 4

Course Description:

Prerequisites: CSE360

CSE445: Computer Vision

Credits: 4

Course Description:

Prerequisites: CSE246

CSE446: ASIC Design Using FPGA

Credits: 4

Course Description:

Prerequisites: CSE345

CSE452: Distributed Systems and Algorithms

Credits: 4

Course Description:

Prerequisites: CSE325

CSE453: Wireless Network

Credits: 4

Course Description:

Prerequisites: CSE405

CSE457: Cellular Networks

Credits: 4

Course Description:

Prerequisites: CSE405

CSE460: Cryptography

Credits: 3

Course Description:

Prerequisites: CSE246

CSE464: Advanced Database System

Credits: 4

Course Description:

Prerequisites: CSE302

CSE471: Compiler Design

Credits: 4

Course Description:

Prerequisites: CSE246

CSE472: Advanced Network Services and Management

Credits: 4

Course Description:

Prerequisites: CSE405

CSE473: Network Security and Systems

Credits: 4

Course Description:

Prerequisites: CSE405

CSE474: Pattern Recognition

Credits: 4

Course Description:

Prerequisites: CSE366

CSE475: Machine Learning

Credits: 4

Course Description:

Prerequisites: CSE366

CSE477: Data Mining

Credits: 4

Course Description:

Prerequisites: CSE366

CSE479: Web Programming

Credits: 4

Course Description:

Prerequisites: CSE302

CSE481: Nature-Inspired Computing

Credits: 4

Course Description:

Prerequisites: CSE246

CSE483: Graph Theory

Credits: 3

Course Description:

Prerequisites: CSE246

CSE484: Computational Geometry

Credits: 3

Course Description:

Prerequisites: CSE246

CSE486: Bioinformatics Algorithms

Credits: 4

Course Description:

Prerequisites: CSE246

CSE487: Computer and Cyber Security

Credits: 3

Course Description:

Prerequisites: CSE405

CSE488: Big Data Analytics

Credits: 4

Course Description:

Prerequisites: CSE302

CSE489: Mobile Programming

Credits: 4

Course Description:

Prerequisites: CSE246

CSE491: VLSI Design

Credits: 4

Course Description:

Prerequisites: CSE345

CSE492: Robotics

Credits: 4

Course Description:

Prerequisites: CSE366

CSE494: Embedded Systems

Credits: 4

Course Description:

Prerequisites: CSE442

CSE495: IT Project Management and Entrepreneurship

Credits: 3

Course Description:

Prerequisites: CSE347

ECO101: Principles of Microeconomics

Credits: 3

Course Description:

Prerequisites:

ECO7101: Principles of Microeconomics

Credits: 3

Course Description:

Prerequisites:

ENG101: Basic English

Credits: 3

Course Description:

Prerequisites:

ENG102: Composition and Communication Skills

Credits: 3

Course Description:

Prerequisites: ENG7101

ENG7101: Basic English

Credits: 3

Course Description:

Prerequisites:

ENG7102: Composition and Communication Skills

Credits: 3

Course Description:

Prerequisites: ENG7101

FIN101: Principles of Finance

Credits: 3

Course Description:

Prerequisites: STA102

GEN201: Bangladesh Studies

Credits: 3

Course Description:

Prerequisites: ENG7101

GEN202: Eastern Culture and Heritage

Credits: 3

Course Description:

Prerequisites:

GEN203: Ecological System and Environment

Credits: 3

Course Description:

Prerequisites:

GEN204: Western Thought

Credits: 3

Course Description:

Prerequisites:

GEN205: Introduction to Psychology

Credits: 3

Course Description:

Prerequisites:

GEN207: Industrial Psychology

Credits: 3

Course Description:

Prerequisites:

GEN208: Introduction to Philosophy

Credits: 3

Course Description:

Prerequisites:

GEN209: Social Psychology

Credits: 3

Course Description:

Prerequisites:

GEN210: International Relations

Credits: 3

Course Description:

Prerequisites: ENG7101

GEN211: Concepts of Journalism & Media Studies

Credits: 3

Course Description:

Prerequisites:

GEN212: Women in Development

Credits: 3

Course Description:

Prerequisites:

GEN213: Introduction to German Language

Credits: 3

Course Description:

Prerequisites:

GEN214: Introduction to Development Studies

Credits: 3

Course Description:

Prerequisites:

GEN215: Introduction to French Language

Credits: 3

Course Description:

Prerequisites:

GEN216: Introduction to Spanish Language

Credits: 3

Course Description:

Prerequisites:

GEN217: Introduction to Chinese Language

Credits: 3

Course Description:

Prerequisites:

GEN218: Introduction to Arabic Language

Credits: 3

Course Description:

Prerequisites:

GEN220: Principles of Public Relations

Credits: 3

Course Description:

Prerequisites:

GEN221: Globalization, Development and Change

Credits: 3

Course Description:

Prerequisites:

GEN223: Contemporary Security Studies in Asia-Pacific

Credits: 3

Course Description:

Prerequisites:

GEN224: Bengali Language

Credits: 3

Course Description:

Prerequisites:

GEN226: Emergence of Bangladesh

Credits: 3

Course Description:

Prerequisites: ENG7102

GEN239: Professional Ethics

Credits: 3

Course Description:

Prerequisites:

GEN7201: Bangladesh Studies

Credits: 3

Course Description:

Prerequisites:

GEN7202: Eastern Culture and Heritage

Credits: 3

Course Description:

Prerequisites:

GEN7203: Ecological System and Environment

Credits: 3

Course Description:

Prerequisites:

GEN7204: Western Thought

Credits: 3

Course Description:

Prerequisites:

GEN7205: Introduction to Psychology

Credits: 3

Course Description:

Prerequisites:

GEN7207: Industrial Psychology

Credits: 3

Course Description:

Prerequisites:

GEN7208: Introduction to Philosophy

Credits: 3

Course Description:

Prerequisites:

GEN7209: Introduction to Social Psychology

Credits: 3

Course Description:

Prerequisites:

GEN7210: International Relation

Credits: 3

Course Description:

Prerequisites:

GEN7211: Concepts of Journalism & Media Studies

Credits: 3

Course Description:

Prerequisites:

GEN7213: Introduction to German Language

Credits: 3

Course Description:

Prerequisites:

GEN7214: Introduction to Development Studies

Credits: 3

Course Description:

Prerequisites:

GEN7215: Introduction to French Language

Credits: 3

Course Description:

Prerequisites:

GEN7216: Introduction to Spanish Language

Credits: 3

Course Description:

Prerequisites:

GEN7217: Introduction to Chinese Language

Credits: 3

Course Description:

Prerequisites:

GEN7218: Introduction to Arabic Language

Credits: 3

Course Description:

Prerequisites:

GEN7220: Principles of Public Relations

Credits: 3

Course Description:

Prerequisites:

GEN7221: Globalization, Development and Change

Credits: 3

Course Description:

Prerequisites:

GEN7223: Contemporary Security Studies in Asia-Pacific

Credits: 3

Course Description:

Prerequisites:

GEN7224: Bangla Language

Credits: 3

Course Description:

Prerequisites:

GEN7226: Emergence of Bangladesh

Credits: 3

Course Description:

Prerequisites: ENG7102

GEN7239: Professional Ethics

Credits: 3

Course Description:

Prerequisites:

MAT101: Differential & Integral Calculus

Credits: 3

Course Description:

Prerequisites:

MAT102: Differential Equations & Special Functions

Credits: 3

Course Description:

Prerequisites: MAT101

MAT104: Co-ordinate Geometry & Vector Analysis

Credits: 3

Course Description:

Prerequisites: MAT101

MAT205: Linear Algebra & Complex Variables

Credits: 3

Course Description:

Prerequisites: MAT102

MGT321: Industrial Management

Credits: 3

Course Description:

Prerequisites: ENG7102

MGT337: Production Operations Management

Credits: 3

Course Description:

Prerequisites: STA102

MKT101: Principles of Marketing

Credits: 3

Course Description:

Prerequisites:

PHY109: Engineering Physics-I

Credits: 4

Course Description:

Prerequisites:

PHY209: Engineering Physics-II

Credits: 3

Course Description:

Prerequisites:

POP201: Health Challenges of Adolescents and Youths

Credits: 3

Course Description:

Prerequisites:

POP202: Introduction to Public Health

Credits: 3

Course Description:

Prerequisites:

POP203: Introduction to Population Studies

Credits: 3

Course Description:

Prerequisites:

POP7201: Health Challenges of Adolescents and Youth

Credits: 3

Course Description:

Prerequisites:

POP7202: Introduction to Public Health

Credits: 3

Course Description:

Prerequisites:

POP7203: Introduction to Population Studies

Credits: 3

Course Description:

Prerequisites:

SOC211: Eastern Culture and Heritage

Credits: 3

Course Description:

Prerequisites:

SOC217: Religion, Ethnicity, Culture and Development in South Asia

Credits: 3

Course Description:

Prerequisites: ENG7101

SOC317: Sociology of Science and Technology

Credits: 3

Course Description:

Prerequisites:

SOC7217: Religion, Ethnicity, Culture and Development in South Asia

Credits: 3

Course Description:

Prerequisites:

STA102: Statistics and Probability

Credits: 3

Course Description:

Prerequisites: MAT101

Course wise Advising Criteria (Depends on Completed Credit) Prerequisites

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
1	ACT101	Financial Accounting	15	140	0	0	
2	BUS231	Business Communication	15	140	0	0	ENG7101
3	BUS321	Business for Engineering and Technology	15	140	0	0	ENG7101
4	CHE109	Engineering Chemistry-I	0	140	0	0	
5	CSE103	Structured Programming	0	140	0	0	
6	CSE106	Discrete Mathematics	0	140	0	0	CSE103
7	CSE110	Object Oriented Programming	10	140	0	0	CSE103
8	CSE200	Computer-Aided Engineering Drawing	15	140	0	0	
9	CSE207	Data Structures	29	140	0	0	CSE110
10	CSE209	Electrical Circuits	20	140	0	0	
11	CSE225	Numerical Methods	90	140	0	0	CSE103
12	CSE246	Algorithms	35	140	0	0	CSE207
13	CSE251	Electronic Circuits	35	140	0	0	CSE209
14	CSE302	Database Systems	45	140	0	0	CSE106
15	CSE313	Theory of Computations	90	140	0	0	CSE246
16	CSE325	Operating Systems	45	140	0	0	CSE207

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
17	CSE345	Digital Logic Design	45	140	0	0	CSE251
18	CSE347	Information System Analysis and Design	45	140	0	0	CSE302
19	CSE350	Data Communications	50	140	0	0	CSE251
20	CSE355	Digital System Design	67	140	0	0	CSE345
21	CSE360	Computer Architecture	50	140	0	0	CSE325
22	CSE366	Artificial Intelligence	65	140	0	0	CSE246
23	CSE400	Capstone Project	95	140	0	0	
24	CSE405	Computer Networks	65	140	0	0	CSE246
25	CSE406	Internet of Things	97	140	0	0	CSE405
26	CSE412	Software Engineering	60	140	0	0	CSE347
27	CSE420	Computer Graphics	85	140	0	0	CSE246
28	CSE422	Simulation and Modeling	97	140	0	0	CSE246
29	CSE423	Software Architecture	97	140	0	0	CSE412
30	CSE428	Human Computer Interactions	97	140	0	0	CSE412
31	CSE430	Software Testing and Quality Assurance	70	140	0	0	CSE412

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
32	CSE432	Digital Signal Processing	85	140	0	0	CSE246,CSE347
33	CSE438	Digital Image Processing	97	140	0	0	CSE246
34	CSE442	Microprocessors and Microcontrollers	85	140	0	0	CSE360
35	CSE452	Distributed Systems and Algorithms	97	140	0	0	CSE325
36	CSE453	Wireless Network	80	140	0	0	CSE405
37	CSE457	Cellular Networks	97	140	0	0	CSE405
38	CSE460	Cryptography	97	140	0	0	CSE246
39	CSE464	Advanced Database System	80	140	0	0	CSE302
40	CSE471	Compiler Design	97	140	0	0	CSE246
41	CSE474	Pattern Recognition	97	140	0	0	CSE366
42	CSE475	Machine Learning	90	140	0	0	CSE366
43	CSE477	Data Mining	90	140	0	0	CSE366
44	CSE479	Web Programming	90	140	0	0	CSE302
45	CSE483	Graph Theory	97	140	0	0	CSE246
46	CSE484	Computational Geometry	97	140	0	0	CSE246
47	CSE486	Bioinformatics Algorithms	97	140	0	0	CSE246
48	CSE487	Computer and Cyber Security	97	140	0	0	CSE405

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
49	CSE489	Mobile Programming	75	140	0	0	CSE246
50	CSE491	VLSI Design	97	140	0	0	CSE345
51	CSE492	Robotics	97	140	0	0	CSE366
52	CSE494	Embedded Systems	97	140	0	0	CSE442
53	ECO101	Principles of Microeconomics					
54	ECO102	Introduction to Macroeconomics					ECO101
55	ENG101	Basic English	0	140	0	0	
56	ENG102	Composition and Communication Skills	3	140	0	0	ENG7101
57	FIN101	Principles of Finance	10	140	0	0	STA102
58	GEN201	Bangladesh Studies	12	140	0	0	ENG7101
59	GEN202	Eastern Culture and Heritage	12	140	0	0	
60	GEN203	Ecological System and Environment	12	140	0	0	
61	GEN204	Western Thought	12	140	0	0	
62	GEN205	Introduction to Psychology	12	140	0	0	
63	GEN207	Industrial Psychology	12	140	0	0	
64	GEN208	Introduction to Philosophy	12	140	0	0	

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
65	GEN209	Social Psychology	12	140	0	0	
66	GEN210	International Relations	12	140	0	0	ENG7101
67	GEN211	Concepts of Journalism & Media Studies	12	140	0	0	
68	GEN212	Women in Development	12	140	0	0	
69	GEN213	Introduction to German Language	12	140	0	0	
70	GEN214	Introduction to Development Studies	12	140	0	0	
71	GEN215	Introduction to French Language	12	140	0	0	
72	GEN216	Introduction to Spanish Language	12	140	0	0	
73	GEN217	Introduction to Chinese Language	12	140	0	0	
74	GEN218	Introduction to Arabic Language	12	140	0	0	
75	GEN220	Principles of Public Relations	12	140	0	0	
76	GEN221	Globalization, Development and Change	12	140	0	0	
77	GEN223	Contemporary Security Studies in Asia-Pacific	12	140	0	0	
78	GEN224	Bengali Language	12	140	0	0	
79	GEN226	Emergence of Bangladesh	12	140	0	0	ENG7102

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
80	GEN239	Professional Ethics	12	140	0	0	
81	MAT101	Differential & Integral Calculus	0	140	0	0	
82	MAT102	Differential Equations & Special Functions	3	140	0	0	MAT101
83	MAT104	Co-ordinate Geometry & Vector Analysis	13	140	0	0	MAT101
84	MAT205	Linear Algebra & Complex Variables	38	140	0	0	MAT102
85	MGT321	Industrial Management	12	140	0	0	ENG7102
86	MGT337	Production Operations Management	12	140	0	0	STA102
87	MKT101	Principles of Marketing	12	140	0	0	
88	PHY109	Engineering Physics-I	3	140	0	0	
89	PHY209	Engineering Physics-II	30	140	0	0	
90	POP201	Health Challenges of Adolescents and Youths	12	150	0	0	
91	POP202	Introduction to Public Health	12	150	0	0	
92	POP203	Introduction to Population Studies	12	150	0	0	
93	SOC211	Eastern Culture and Heritage	12				

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
94	SOC217	Religion, Ethnicity, Culture and Development in South Asia	12	140	0	0	ENG7101
95	SOC317	Sociology of Science and Technology	12	140	0	0	
96	STA102	Statistics and Probability	26	140	0	0	MAT101
97	CSE407	Green Computing	95	140	0	0	CSE405
98	CSE495	IT Project Management and Entrepreneurship	110	140	0	0	CSE347
99	CSE326	Algorithms	60	140	0	0	CSE207
100	CSE303	Statistics for Data Science	60	140	0	0	STA102
101	CSE445	Computer Vision	97	140	0	0	CSE246
102	CSE481	Nature-Inspired Computing	97	140	0	0	CSE246
103	CSE488	Big Data Analytics	97	140	0	0	CSE302
104	CSE472	Advanced Network Services and Management	97	140	0	0	CSE405
105	CSE473	Network Security and Systems	97	140	0	0	CSE405
106	CSE446	ASIC Design Using FPGA	97	140	0	0	CSE345
107	CSE400A	Capstone Project (Part 1 of 3)	95	140	0	0	
108	CSE400B	Capstone Project (Part 2 of 3)	95	140	0	0	CSE400A

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
109	CSE400C	Capstone Project (Part 3 of 3)	95	140	0	0	CSE400B
110	ENG7101	Basic English	0	140			
111	ENG7102	Composition and Communication Skills	3	140			ENG7101
112	GEN7226	Emergence of Bangladesh	12	140			ENG7102
113	GEN7201	Bangladesh Studies					
114	GEN7205	Introduction to Psychology					
115	GEN7207	Industrial Psychology					
116	GEN7239	Professional Ethics					
117	ECO7101	Principles of Microeconomics					
118	GEN7204	Western Thought					
119	GEN7210	International Relation					
120	GEN7211	Concepts of Journalism & Media Studies					
121	GEN7213	Introduction to German Language					
122	GEN7202	Eastern Culture and Heritage					
123	GEN7208	Introduction to Philosophy					

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
124	GEN7209	Introduction to Social Psychology					
125	GEN7214	Introduction to Development Studies					
126	GEN7220	Principles of Public Relations					
127	GEN7215	Introduction to French Language					
128	GEN7216	Introduction to Spanish Language					
129	GEN7217	Introduction to Chinese Language					
130	GEN7218	Introduction to Arabic Language					
131	GEN7221	Globalization, Development and Change					
132	GEN7223	Contemporary Security Studies in Asia-Pacific					
133	GEN7224	Bangla Language					
134	POP7201	Health Challenges of Adolescents and Youth					
135	POP7202	Introduction to Public Health					
136	POP7203	Introduction to Population Studies					
137	SOC7217	Religion, Ethnicity, Culture and Development in South Asia					

SL	Course Code	Course Name	Min. Credit	Max. Credit	Min. CGPA	Max. CGPA	Course Prerequisites
138	GEN7203	Ecological System and Environment					